

SECTION B

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
8				Indicative content use insulated container add known volume of HCl / known concentration excess to ensure that all CaO reacts measure initial temperature of HCl add known mass of CaO and stir well for rapid and complete reaction measure temperature change at intervals plot temperature change over time and extrapolate lines determine temperature increase repeat all steps using Ca(OH)₂ calculate number of moles of CaO / Ca(OH)₂ reacted $n = \frac{mass}{M_r}$ calculate enthalpy change ΔH_2 and ΔH_3 $\Delta H = - \frac{mc\Delta T}{n}$ use Hess's cycle to calculate ΔH_1 $\Delta H_1 = \Delta H_2 - \Delta H_3$	4		2	6	2	4